Vision Aided Navigation - Exercise 4

Student: Shay Morad

GitHub Repository: https://github.com/ShayMorad/3D-Computer-Vision-SLAM

(Note: Code is located on the main branch. Collaborator invites have been sent to course teachers).

Part 4.2: Global Tracking Statistics

Following the serialization of the 3,300-frame sequence, the database was queried to evaluate the global stability of the temporal tracking pipeline. To ensure statistical relevance, all trivial tracks (features successfully matched in stereo but failing to persist into the next temporal frame; length = 1) were excluded from the dataset.

- **Total number of tracks:** 360,321
- **Number of frames:** 3,300
- **Mean track length:** 4.75 frames
- **Maximum track length:** 104 frames
- **Minimum track length:** 2 frames
- **Mean number of frame links:** 518.26 tracks per image

```
--- Q 4.2: Tracking Statistics ---
Total number of tracks (length > 1): 360321
Number of frames: 3300
Mean track length: 4.75 frames
Maximum track length: 104 frames
Minimum track length: 2 frames
Mean number of frame links (tracks per image): 518.26
```

Code Reference: ex4.py, Lines 90-117

Part 4.3: Visual Track Validation (20x20 Crop)

To visually validate the integrity of the database index mapping, a track of length 10 was isolated. The feature's coordinates were queried from the TrackingDB, and a 20x20 pixel bounding box was cropped around the location across all consecutive frames. As demonstrated in the figure, the red tracking marker remains rigidly locked onto the exact same physical geometric corner despite the forward translation and perspective shift of the vehicle. This confirms that the database indexing flawlessly preserves the relational associations of the temporal matches.

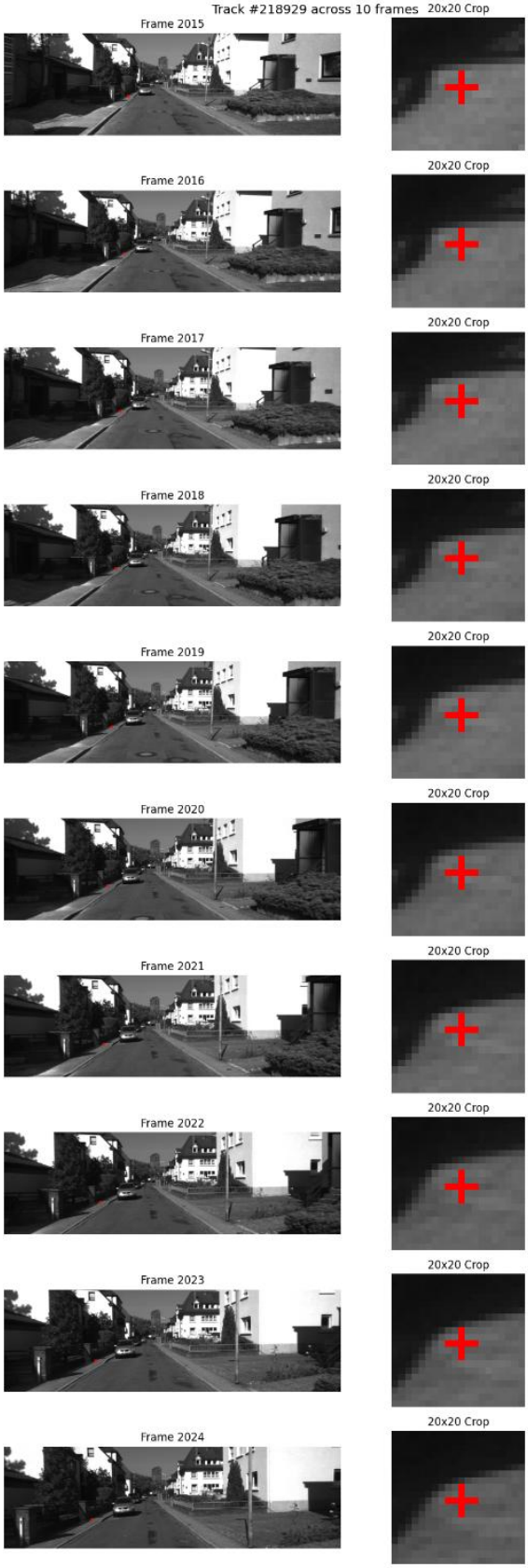


Figure 1: 20x20 Track #218929 across 10 frames

Code Reference: ex4.py, Lines 127.

Part 4.4: Frame Connectivity

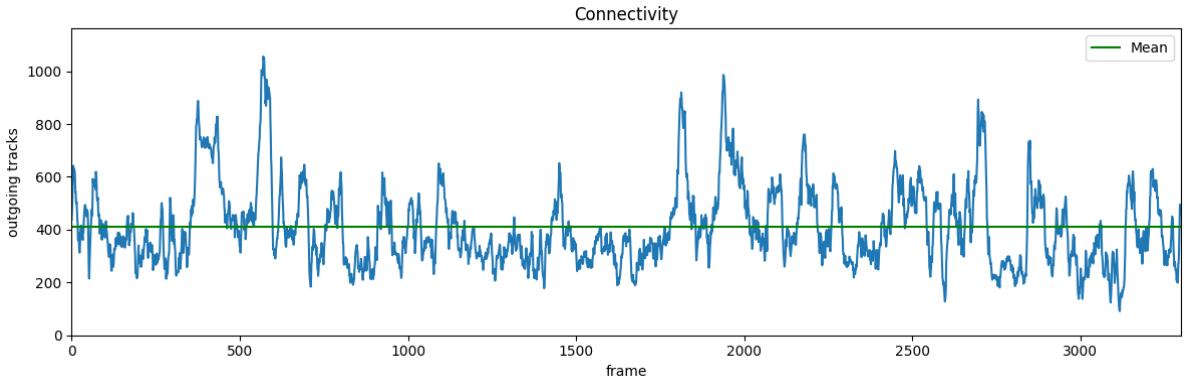


Figure 2: Connectivity Plot
The connectivity graph measures the number of outgoing tracks (features successfully linked to the subsequent frame) over the sequence.

Code Reference: ex4.py, Line 143.

Analysis: The system maintains a highly stable mean connectivity of approximately 415 tracks. Crucially, the outgoing track count never drops to zero, ensuring the pipeline never loses its spatial reference (which would result in catastrophic trajectory failure). The visible deep troughs (e.g., around frames 900 and 1,800) naturally correlate with sharp turns or rapid rotational movements in the KITTI trajectory, where the camera's field of view shifts too rapidly for features to persist across multiple frames.

Part 4.5: Inlier Percentage per Frame

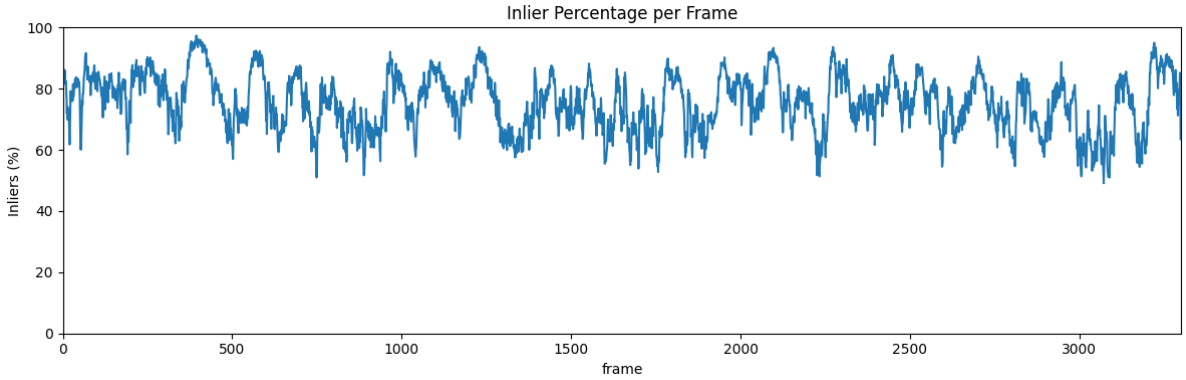


Figure 3: Inlier Percentage Plot

Code Reference: ex4.py, Line 161.

Analysis: The inlier percentage fluctuates healthily between 50% and 95%. This high retention rate proves that the AKAZE detector successfully extracts structurally stable environmental features. The discarded percentage (outliers) represents the successful

mathematical rejection of dynamic objects (e.g., oncoming traffic), transient shadows, and mismatched descriptors.

Part 4.6: Track Length Distribution

A histogram (plotted on a logarithmic Y-axis) was generated to visualize the lifespan of all valid tracks (length > 1):

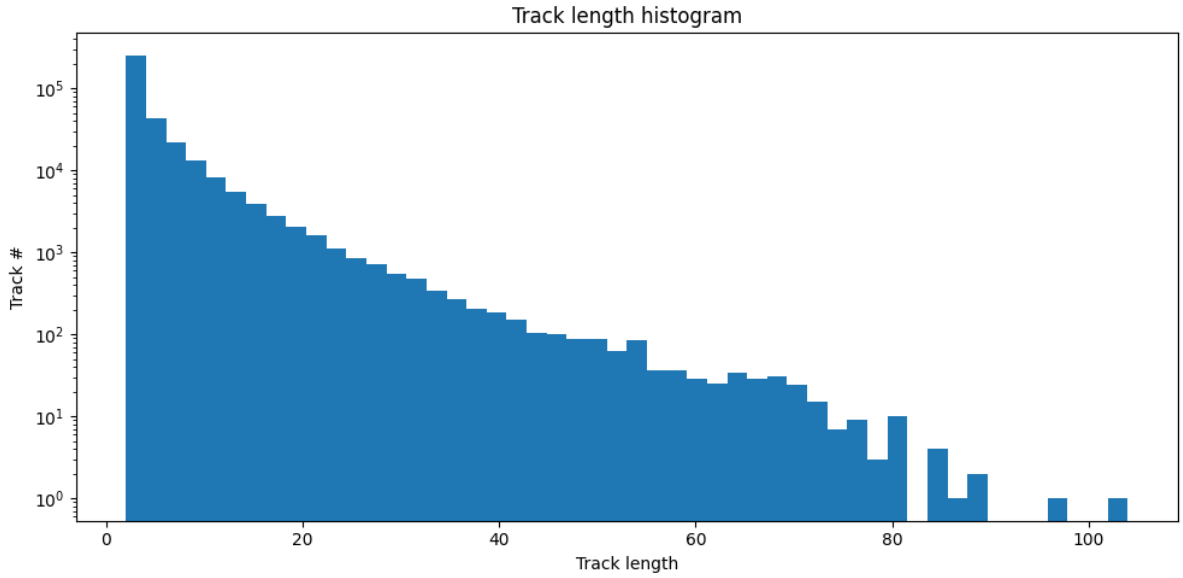


Figure 4: Track Length Histogram

Code Reference: ex4.py, Line 170.

Analysis: The distribution follows a textbook exponential decay. The vast majority of tracks persist for a short duration; these represent physical objects in the immediate foreground (like road markings or passing parked cars) that quickly translate out of the camera's field of view. However, the system successfully maintains a vital "long tail" of tracks spanning 50 to over 100 frames. These highly persistent features represent distant, stable anchors (such as horizon lines or tall architecture) that provide the long-term orientation constraints required for minimizing drift in backend optimization.

Part 4.7: Reprojection Analysis and Triangulation Theory

A track of length >= 10 was isolated, and its 3D world coordinate was triangulated using the Ground Truth matrices from its final frame. This coordinate was projected backward through the preceding frames, revealing a gradual accumulation of sub-pixel tracking error (drift) as the distance from the reference frame increased:

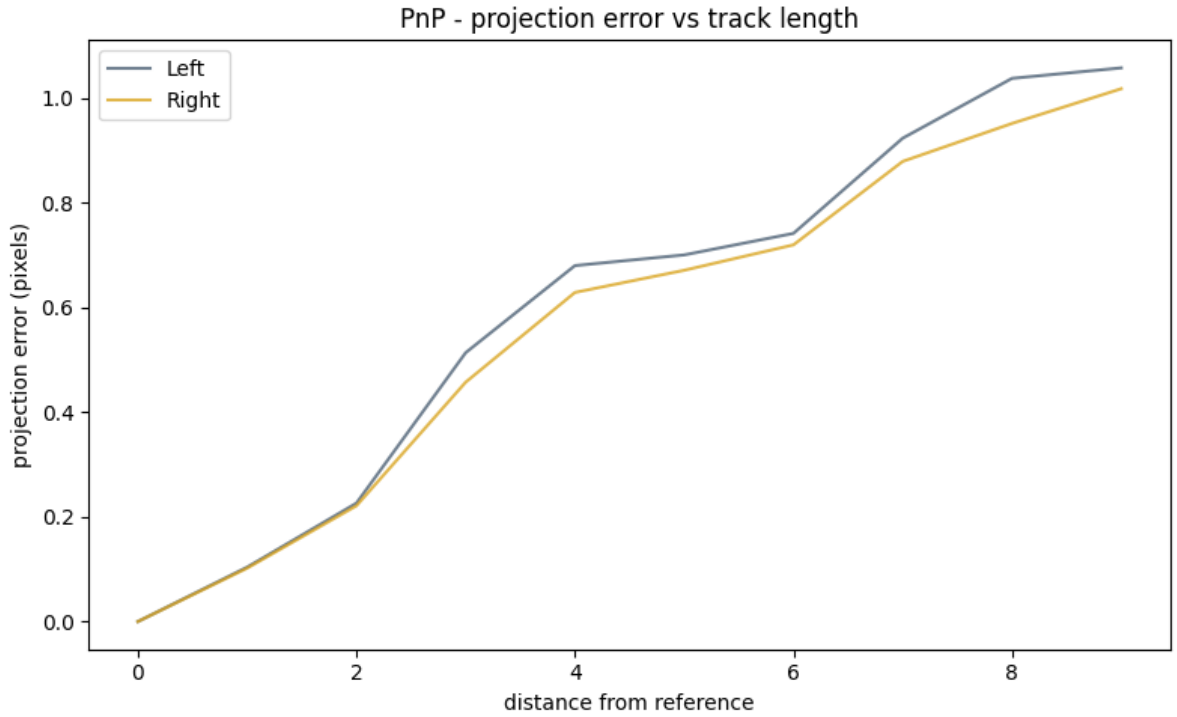


Figure 5: Projection Error vs Track Length

Code Reference: ex4.py, Line 169.

- **Theoretical Question:** *What would happen if we triangulated from the first frame instead of the last? How would you expect the reprojection error to change? Why?*

- If triangulation originated from the first frame rather than the last, the overall magnitude of the reprojection error projected forwards would be significantly larger.

  This occurs due to the mathematical relationship between depth (Z) and stereo triangulation uncertainty. Because the cameras have a fixed physical baseline, depth uncertainty scales quadratically with distance.
  At the first frame, the tracked feature is geographically far from the vehicle, resulting in high triangulation noise. As the vehicle drives toward the feature, depth decreases, maximizing the signal-to-noise ratio at the *last* frame. Therefore, triangulating from the last frame provides a highly accurate 3D spatial anchor. Triangulating from the first frame anchors the sequence to a noisy estimate, causing the forward-projected error curve to scale drastically.

Q 4.8

1. Proof of mean and covariance:

By the linearity of Expectation:

$$\mathbb{E}[y] = \mathbb{E}[Ax+b] = A\mathbb{E}[x]+b = A\mu_x + b = \mu_y$$

Covariance:

$$\Sigma_y = \mathbb{E}[(y-\mu_y)(y-\mu_y)^T] = \mathbb{E}\left[(Ax+b-(A\mu_x+b))(Ax+b-(A\mu_x+b))^T\right] =$$

$$= \mathbb{E}\left[A(x-\mu_x)(A(x-\mu_x))^T\right] = A\cdot\mathbb{E}[(x-\mu_x)(x-\mu_x)^T]\cdot A^T =$$

$$= A\cdot\Sigma_x\cdot A^T$$

2. Proof of probability density function:

$y = Ax+b \rightarrow g(x) := Ax+b \rightarrow g^{-1}(y) = x = A^{-1}(y-b), \quad |Jg^{-1}| = \frac{1}{|A|}.$

$$f_y(y) = \frac{1}{\sqrt{(2\pi)^k\cdot|\Sigma_x|\cdot|A|^2}}\cdot\exp\left(-\frac{1}{2}\cdot\|A^{-1}(y-b)-\mu_x\|^2_{\Sigma_x}\right)$$

$$\sqrt{|\Sigma_x|\cdot|A|^2} = \sqrt{|A\cdot\Sigma_x\cdot A^T|} = \sqrt{|\Sigma_y|}$$

The inner Mahalahobis term simplifies to: $A^{-1}(y-b)-A^{-1}A\mu_x = A^{-1}(y-\mu_y)$

Applying the Mahalahobis norm property with $M=A^{-1}$, $u=y-\mu_y$:

$$\|A^{-1}(y-\mu_y)\|^2_{\Sigma_x} = \|y-\mu_y\|^2_{(A^{-1})^{-1}\cdot\Sigma_x\cdot(A^{-1})^T} = \|y-\mu_y\|^2_{A\Sigma_xA^T} = \|y-\mu_y\|^2_{\Sigma_y}$$

$$\Rightarrow f_y(y) = \frac{1}{\sqrt{(2\pi)^k\cdot|\Sigma_y|}}\cdot\exp\left(-\frac{1}{2}\cdot\|y-\mu_y\|^2_{\Sigma_y}\right), \text{ Thus } y\sim N(A\mu_x+b, A\Sigma_xA^T)$$